SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2*:* Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: **5**

Total **Marks**:50

Annexure A

ANALYTICAL **PROBLEM SOLVING**

Code of Conduct:

I Samuel V

(192412237)

(Name/Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Land

V

2nd **SSD Slot**
AURA
RGB
AURA RGB
D

Annexure A

ANALYTICAL PROBLEM SOLVING

A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid

host-IP

range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

34 An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR/25, Sales → /26, IT → /27, and Admin /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a /27 **subnet** mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the

number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:**0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

accurately Selects
most appropriate
method/formu
Selects
Method Selection
20%
la with strong
justification
correct method with minor errors

Solution Process
25%
Inappropriate or
partially correct method

Logical,
systematic
steps with
correct
approach
throughout
All
Mostly
correct steps
with minor
mistakes
Disorganized
steps; multiple
errors

Accuracy *of*
calculations
20%
Calculation
accurate; correct
final
Minor calculation
errors
answer
Clearly
interprets
Interpretation *of*

Results

results with

15%

units and

engineering

relevance

Unable to

select method

No clear

process

Frequent

errors; incorrect final

answer

No valid

calculations

Basic

interpretation

with minor

omissions

Limited or

unclear

interpretation

No

interpretation

given :-

18)

Network Address: 192.168. 20.0/24

Lab A ->/26

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24.07.2026

Lab B/27

Lab c→/28

(302- Prefix)

Formula :-

Usable Host = &

2

2

Lab A (126)

Subnet Mask: 255.255.255.192

Network Address: 192.168.20.65

First Host: 192.168. 20.65

Last Host: 192.168.20.94.

Broadkast Address : 192.168.20.95

Lab B

(127)

Subnet: 255.255.255.224

Mask

Network 192.168.20.64/27

Address

Last

Host

192.168.20.94

Total Address =

302-026

22

2) 64

Usable

Host

64-2=62

Lab (128)

Sa 255.255.255.240

192.168.20.97

192.168.20.110

Broadlast :

192.168.20.95

Address

Total Addresses: 3-28

192.168.20.111

d

16

32

usable Host:

16-2=39

14

Given

Network

Address: 172.16.0.0/16

HR/25

Sales /26

IT → /27

Admin -/28

Usable Hosts = 2

(30- Prefix)

2

Dept

Prefix

Subnet Mask

Network

Broadcast Usable

Hosts

Address

Address

HR

125

255.255.255.168.

172.16.0.0

172.16.0.127

1226

Salas

126

255.255.255 192

172.16.0.128 172.16.0.191

622

IT

127

255.255.255. 224 172.16.0.1.92

172.16.0.223

30

min

128

255.255.255.240

172.16.0.224

172 16.0.239

14